COURSE-BASED PROJECT DOCUMENTATION | DOC-13

Tkinter GUI Architecture

VS Code-Style UI, Command Palette, & Line Number Synchronization

1. Graphical User Interface Overview

The GUI application (gui/app.py) is a modern, VS Code-inspired developer tool built entirely with Python standard libraries. It provides an intuitive environment for writing, opening, and verifying Java 21 source code.

2. Core GUI Subsystems

- Editor with Synchronized Gutter: LineNumberText wraps a text area and left canvas displaying right-aligned line numbers.
- Dark Developer Theme: Custom palette (#1e1e1e editor, #252526 panels, #007acc accent, #5a1d1d error tint).
- Command Palette (Ctrl+Shift+P / F1): Floating quick-pick overlay with real-time fuzzy search across all commands.
- Status Bar: Compact bar displaying status, cursor position (Ln X, Col Y), encoding (UTF-8), and engine status.
- Output Panel: 4-line terminal log recording timestamped execution history ($ check Sample.java).

3. Keyboard Shortcuts Reference

Shortcut Key	Action Description	Palette Command
Ctrl + Shift + P / F1	Open Command Palette quick-picker	Open Command Palette
Ctrl + Enter / F5	Check Java Syntax	Check Syntax
Ctrl + O	Open .java source file from disk	Open File...
Ctrl + S	Save current editor text to file	Save File
Ctrl + L	Clear editor buffer and status	Clear Editor
Escape	Dismiss Command Palette overlay	Close Palette

4. Visual Error-Line Highlighting

When an error on line N is returned, the editor tags the line range ('N.0' to 'N.end+1c') with the error_line tag (soft red background #5a1d1d), colors gutter number N red, and scrolls line N into focus using self.editor.text.see('N.0').